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ARROW-PRATT MEASURES OF RISK AVERSION: THE MULTIVARIATE CASE*

BY HAIM LEVY AND AZRIEL LEVY¹

Arrow-Pratt measures of risk aversion have been defined for the univariate case. For utility functions having the same ordinal preferences, we extend Arrow's probability premium index to the multivariate case and obtain a unique solution which can be employed to risk-aversion comparison analysis. We also extend Duncan's definition of the risk premium vector and show that it can be employed in comparative risk aversion once we confine ourselves to the same preference ordering. Hence, we end up with two multivariate risk indexes which are parallel to the Arrow and Pratt univariate indexes.

1. INTRODUCTION

Arrow (1971) and Pratt (1964) have developed the theory of risk aversion for the univariate utility function $V(x)$. Moreover, they compare the degree of risk aversion which characterizes two distinct utility functions $V^1(x)$ and $V^2(x)$. The extension of the Arrow-Pratt theory of risk aversion to the multivariate case raises certain issues that do not exist in the univariate case. In particular, any comparison of the degree of risk aversion of two multivariate utility functions $u^1(x_1, \dots, x_n)$ and $u^2(x_1, \dots, x_n)$ (where x is a vector of various commodities) is impossible unless some constraints are imposed on the utility functions under consideration. Kihlstrom and Mirman (1974) argue that a prerequisite for the comparison of attitudes towards risk is that the cardinal utilities being compared represent the same ordinal preferences. Nevertheless, even for the same preference ordering, the multivariate risk premium, as Duncan (1977) points out, is not unique, which further complicates the comparison of the risk premium of different individuals.

Extending Pratt's approximation of the univariate risk premium, Duncan (1977) develops a multivariate risk aversion matrix and reveals its relation to an approximate (in the "small") multivariate risk premium. This derivation holds for any utility function. Namely, Duncan does not confine himself to the same preference ordering; hence, a risk aversion comparison analysis is meaningless.

In this paper, we provide a generalization of Arrow's (1971) univariate gamble to the multivariate case. In a manner analogous to Arrow's derivation we derive the probability premium of the gamble. The advantage of this probability premium approach is that it provides a unique solution and, therefore, an unambiguous criterion for the comparison of the risk aversion of various individuals.

We next show that this advantage is of a technical nature only. We extend Duncan's work and show that for utility functions which belong to the same

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preference ordering, a risk aversion comparison is possible in spite of the fact that the risk premium vector is not unique.

In Section 2, we review the results of previous studies which are relevant to this paper. In Section 3, we introduce the concept of a *single vector risk* and derive the probability premium. In Section 4, we show the relation to Duncan's work. Concluding remarks are given in Section 5.

2. DEFINITIONS AND REVIEW

The theory of risk aversion and the application of the various risk aversion measures to situations involving risk was initially developed by Arrow (1971) and Pratt (1964).² Given a twice continuously differentiable utility function $V(x)$, they define an absolute and a relative risk aversion measure. Arrow deals with a gamble which is a special case of Pratt's gamble.

The extension of the theory of risk aversion to multivariate utility functions and multivariate risk raises some difficulties that do not arise in the univariate case. In the multivariate case, there exists an infinite set of (ordinal) preference orderings.³ Each such preference ordering, in turn, implies an infinite number of cardinal utility functions. Kihlstrom and Mirman (1974) have observed that a prerequisite for the comparison of attitudes towards risk is that the decision makers being compared have identical ordinal preferences over the commodity space.⁴

Pursuing a somewhat different path, Duncan (1977) develops a matrix measure of *local* risk aversion which is related to the multivariate risk premium. Let $\underline{z}$ be a multivariate risk and $u(\underline{x})$ some utility function. A risk premium vector $\pi(\underline{x}, \underline{z})$ is defined as the solution to the following equation:

$$(1) \quad u(\underline{x} + E \underline{z} - \pi) = Eu(\underline{x} + \underline{z})$$

where $\underline{x}$ is constant.

3. MULTIVARIATE PROBABILITY PREMIUM

In this section, we extend Arrow's probability premium to the multivariate case. Our work is a generalization of Arrow's (1971) univariate results, just as Duncan's (1977) work is a generalization of the results of Pratt (1964). However, in contrast with the univariate case, we shall see that the relation between the results of this paper and Duncan's results is not straightforward. Moreover, we show that our definition holds in the small as well as in the large.

² Ross (1981) developed stronger measures of risk aversion that prove useful in cases where there is no risk free situation.

³ Each cardinal utility function belonging to some preference ordering can be obtained as a monotonic transformation of any other utility function that represents the preferences. Thus, we can define the preference ordering by some basic utility functions $u(\underline{x})$ and all its monotonically increasing transformations.

⁴ In the univariate case, this restriction does not hold, since there obviously is only one preference ordering.

We first define the concept of a *single vector risk*, which is analogous to Arrow's bet (Arrow 1971, p. 94):

DEFINITION 1. A single vector risk $\underline{z}$ is defined as

$$\underline{z} = \begin{cases} \underline{h} & \text{with probability } P \\ -\underline{h} & \text{with probability } 1 - P \end{cases}$$

where $\underline{h}' = (h_1, \dots, h_n) \in R_n$ is some vector. ($\underline{h}'$ denotes the transpose of the vector $\underline{h}$.)

We assume that an individual with initial consumption basket $\underline{x}$ and with a *twice continuously differentiable* utility function $u(\underline{x})$ (assumed to be increasing in each argument) is confronted with a single vector risk $\underline{z}$. The vector $\underline{h}$ can be chosen arbitrarily as long as the following inequality holds,

$$(2) \quad u(\underline{x} + \underline{h}) > u(\underline{x}) > u(\underline{x} - \underline{h})$$

($\underline{h}$ can be small or large).

Condition (2) assures that receiving the vector $\underline{h}$ is preferred to remaining with the initial basket $\underline{x}$, which is preferred to losing the vector $\underline{h}$ (this condition obviously holds if $h_i \geq 0$, $i = 1, \dots, n$, where h_i denotes the i th component of the vector $\underline{h}$).

Due to Jensen's inequality, any risk averter will reject the gamble $\underline{z}$ if $P = 1/2$, i.e., if $\underline{z}$ is a fair gamble. As in Arrow's (1971) analysis in the univariate case (assuming condition (2) holds), we can find the probability P under which the individual is indifferent between receiving the bet $\underline{z}$ and remaining with the initial wealth $\underline{x}$.

Equating the expected utility of the gamble and the utility of having the sure vector $\underline{x}$ we get,

$$(3) \quad u(\underline{x}) = Pu(\underline{x} + \underline{h}) + (1 - P)u(\underline{x} - \underline{h}).$$

If the individual is risk averse then,

$$(4) \quad P > \frac{1}{2}.$$

Theorem 1 below is a multivariate generalization of the univariate results of Arrow (1971). The theorem proves the relationship between the probability premium and the concept of being more risk averse.

THEOREM 1. Let $u^1(\underline{x})$ and $u^2(\underline{x})$ be two utility functions, where $u^1(\underline{x})$ is a monotonic transformation of $u^2(\underline{x})$. Let $\underline{h}$ be some single vector risk and P_1, P_2 be the probabilities associated with $u^1(\underline{x})$ and $u^2(\underline{x})$, respectively (equation (3)). The following two conditions are equivalent:

(a) $P_1 \geq P_2$ for every $\underline{x}$ and every $\underline{h}$, satisfying the inequalities in (2).

(b) $u^1(\underline{x})$ is more risk averse than $u^2(\underline{x})$ according to Kihlstrom and Mirman's definition.

Therefore, the larger P the more risk averse is the individual.

PROOF.

I. We first prove that (b) implies (a). It is given that

$$(5) \quad u^1(\underline{x}) = k(u^2(\underline{x}))$$

where $k'(\cdot) > 0$, $k''(\cdot) \leq 0$.

We have to prove that

$$(6) \quad P_1 \geq P_2$$

where P_1 , P_2 are the probability premiums of u^1 and u^2 , obtained from (3). Inequality (6) holds if,

$$(7) \quad 1/P_1 - 1 \leq 1/P_2 - 1.$$

From equation (3) we solve for P_1 (substituting P_1 for P and u^1 for u):

$$P_1 = [u^1(\underline{x}) - u^1(\underline{x} - \underline{h})]/[u^1(\underline{x} + \underline{h}) - u^1(\underline{x} - \underline{h})]$$

$$(8a) \quad 1/P_1 - 1 = [u^1(\underline{x} + \underline{h}) - u^1(\underline{x})]/[u^1(\underline{x}) - u^1(\underline{x} - \underline{h})].$$

Similarly,

$$(8b) \quad 1/P_2 - 1 = [u^2(\underline{x} + \underline{h}) - u^2(\underline{x})]/[u^2(\underline{x}) - u^2(\underline{x} - \underline{h})].$$

For simplicity, denote,

$$(9a) \quad W_1 = u^2(\underline{x} - \underline{h})$$

$$(9b) \quad t = u^2(\underline{x})$$

$$(9c) \quad W_3 = u^2(\underline{x} + \underline{h})$$

where, $W_1 < t < W_3$.

From (5), (8a), (8b), and (9), we find that condition (7) is equivalent to,

$$(10) \quad [k(W_3) - k(t)]/[k(t) - k(W_1)] \leq [W_3 - t]/[t - W_1].$$

This condition is equivalent to

$$(11) \quad [k(W_3) - k(t)]/W_3 - t \leq [k(t) - k(W_1)]/[t - W_1].$$

The assumption $k''(\cdot) \leq 0$ implies the validity of (11) (or (7), which is equivalent of it).

II. We next prove that (a) implies (b). It is given that $P_1 \geq P_2$, for every $\underline{x}$, $\underline{h}$. We have to prove that,

$$(12) \quad u^1(\underline{x}) = k(u^2(\underline{x}))$$

where $k''(\cdot) \leq 0$.

We show in part I that, if $P_1 \geq P_2$ (see (6)) then (11) holds, for any $\underline{x}$, $\underline{h}$. We will show that for any s , t , (such that $s < t$),

$$(13) \quad k'(s) \geq k'(t)$$

where k' is the first derivative of $k()$.

Define t as in (9b), and take $\underline{h} > 0$ and small. By definition,

$$(14) \quad [k(W_3) - k(t)]/[W_3 - t] = \frac{[k(u^2(\underline{x} + \underline{h})) - k(u^2(\underline{x}))]}{u^2(\underline{x} + \underline{h}) - u^2(\underline{x})}.$$

Substitute $\underline{x} - \underline{h}$ instead of $\underline{x}$ in (9a), (9b), (9c), and (11), and obtain

$$(15) \quad \frac{k(u^2(\underline{x} - \underline{h})) - k(u^2(\underline{x} - 2\underline{h}))}{u^2(\underline{x} - \underline{h}) - u^2(\underline{x} - 2\underline{h})} \geq \frac{k(u^2(\underline{x})) - k(u^2(\underline{x} - \underline{h}))}{u^2(\underline{x}) - u^2(\underline{x} - \underline{h})}.$$

We can continue this process by substituting $\underline{x} - (n - 1)\underline{h}$ instead of $\underline{x}$ ($n = 1, 2, \dots$). We finally get,

There exists n (depending on $\underline{h}$) such that,

$$s \in [u^2(\underline{x} - (n - 1)\underline{h}), u^2(\underline{x} - n\underline{h})].$$

Let the norm of $\underline{h}$ go to zero. The limit of the left-hand side of (16) is $k'(s)$, and the limit of the right-hand side is $k'(t)$. Thus, we have found that,

$$(17) \quad k'(s) \geq k'(t).$$

Note that the results of this section break down when the utility functions being compared do not belong to the same preference ordering.

4. RELATION TO THE WORK OF DUNCAN

The gamble defined in Section 3 is a special case of the general risk introduced by Duncan, just as in the univariate case Arrow's gamble is a special case of the general risk introduced by Pratt. However, while in the univariate case both Arrow's probability premium and Pratt's premium depend solely on the absolute risk aversion function, the relationship between Duncan's results and those of Section 3 of this paper is not so obvious.

It is the purpose of this section to show this relationship. To be able to compare the risk aversion of the two utility functions in Duncan's framework, we must limit ourselves to utility functions belonging to the same preference ordering. We show in Theorem 2 below that for such utility functions, one can provide economic meaning to the term "more risk averse" in Duncan's framework. Moreover, we show that Duncan's extension holds in the small as well as in the large.⁵

⁵ See Paroush (1975) proposition 3 for a proof of a special case of Theorem 2.

THEOREM 2. Let $u^1(\underline{x})$ and $u^2(\underline{x})$ be two utility functions representing the same preferences. Let $\underline{z}$ be some risk, small or large, and $\underline{\pi}^1, \underline{\pi}^2$ be the risk premium (a vector, see equation (1)) associated with the risk, pertaining to u^1 and u^2 , respectively. The following two conditions are equivalent:

(a) $u^1(\underline{x})$ is more risk averse than $u^2(\underline{x})$ according to Kihlstrom and Mirman's definition.

(b) $u^1(\underline{x} - \underline{\pi}^2) \geq u^1(\underline{x} - \underline{\pi}^1)$, for all $\underline{x}$ and all risks $\underline{z}$.

PROOF.

I. We first prove that (a) implies (b).

Assume that $k(\cdot)$ is concave (i.e., $k''(\cdot) \leq 0$). We have to prove that $u^1(\underline{x} - \underline{\pi}^2) \geq u^1(\underline{x} - \underline{\pi}^1)$. $\underline{\pi}^2$, the risk premium of $u^2(\underline{x})$, given some risk $\underline{z}$, is obtained from the following equation,

$$(18) \quad u^2(\underline{x} - \underline{\pi}^2) = Eu^2(\underline{x} + \underline{z}),$$

(for simplicity only, we assume $E\underline{z} = 0$). Since $k(\cdot)$ is concave, and due to Jensen's inequality, the following inequality must hold,

$$(19) \quad u^1(\underline{x} - \underline{\pi}^2) = k(u^2(\underline{x} - \underline{\pi}^2)) = k[Eu^2(\underline{x} + \underline{z})] \geq Ek(u^2(\underline{x} + \underline{z})) = Eu^1(\underline{x} + \underline{z}).$$

Likewise, $\underline{\pi}^1$ the risk premium of $u^1(\underline{x})$ given the risk $\underline{z}$, is obtained from the following equation,

$$(20) \quad u^1(\underline{x} - \underline{\pi}^1) = Eu^1(\underline{x} + \underline{z}).$$

Comparing (20) with (19), we obtain the following inequality,

$$u^1(\underline{x} - \underline{\pi}^2) \geq u^1(\underline{x} - \underline{\pi}^1)$$

which completes part I of the proof.

II. We next prove that (b) implies (a). Assume that $u^1(\underline{x} - \underline{\pi}^2) \geq u^1(\underline{x} - \underline{\pi}^1)$, for every $\underline{x}$ and $\underline{z}$ (where $\underline{z}$ is a random vector). We have to prove that $k''(\cdot) \leq 0$.

From the above assumption and the definition of $u^2(\cdot)$ (see (5)), we have

$$(21) \quad u^1(\underline{x} - \underline{\pi}^1) = Eu^1(\underline{x} + \underline{z}) = E\{k(u^2(\underline{x} + \underline{z}))\} \leq u^1(\underline{x} - \underline{\pi}^2) \\ = k(u^2(\underline{x} - \underline{\pi}^2)) = k[Eu^2(\underline{x} + \underline{z})].$$

Thus, from (21) we have,

$$(22) \quad E\{k(u^2(\underline{x} + \underline{z}))\} \leq k[E\{u^2(\underline{x} + \underline{z})\}]$$

for every $\underline{x}$ and every risk $\underline{z}$.

We have to prove that $k(\cdot)$ is concave. We will show that for any t, s , and $0 \leq \lambda \leq 1$

$$(23) \quad \lambda k(t) + (1 - \lambda)k(s) \leq k(\lambda t + (1 - \lambda)s).$$

For any t, s , chosen from the range of u^2 , we can find some $\underline{x}, \underline{y}^1$, and $\underline{y}^2$, such that $t = u^2(\underline{x} + \underline{y}^1)$; $s = u^2(\underline{x} + \underline{y}^2)$. Define the gamble $\underline{z}$,

$$(24) \quad \underline{z} = \begin{cases} y^1 & \text{with probability } \lambda \\ y^2 & \text{with probability } 1 - \lambda \end{cases}.$$

Substituting (24) into (22) gives (23), which, together with the continuity assumption, proves concaveness.

The economic interpretation of Theorem 2 is that the more risk averse the individual, the more he is willing to give up in terms of utility, in order to avoid the risk. To obtain this result, we had to turn back to the original definition of the multivariate risk premium (equation (1)) and could not receive it directly from Duncan's approximation.

5. CONCLUDING REMARKS

The univariate concept of probability risk premium has been extended to the multivariate case, for utility functions belonging to the same preference ordering. This analysis constitutes a generalization of Arrow's univariate probability risk premium. We prove that if and only if the probability risk premium corresponding to utility function $u^1(\underline{x})$ is greater than the probability risk premium corresponding to $u^2(\underline{x})$ for all $\underline{x}$, then there exists a monotonically increasing and concave transformation k such that $u^1(\underline{x}) = k(u^2(\underline{x}))$. These results are independent of the chosen gamble vector $\underline{h}$, which could be "small" or "large."

For utility functions which belong to the same preference ordering, we circumvent the nonuniqueness of the vector of risk premiums and show that Duncan's framework can be employed in the comparison of the risk aversion of two agents. Thus, we establish a criterion which enables comparative risk aversion in the Pratt and Duncan framework as well.

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